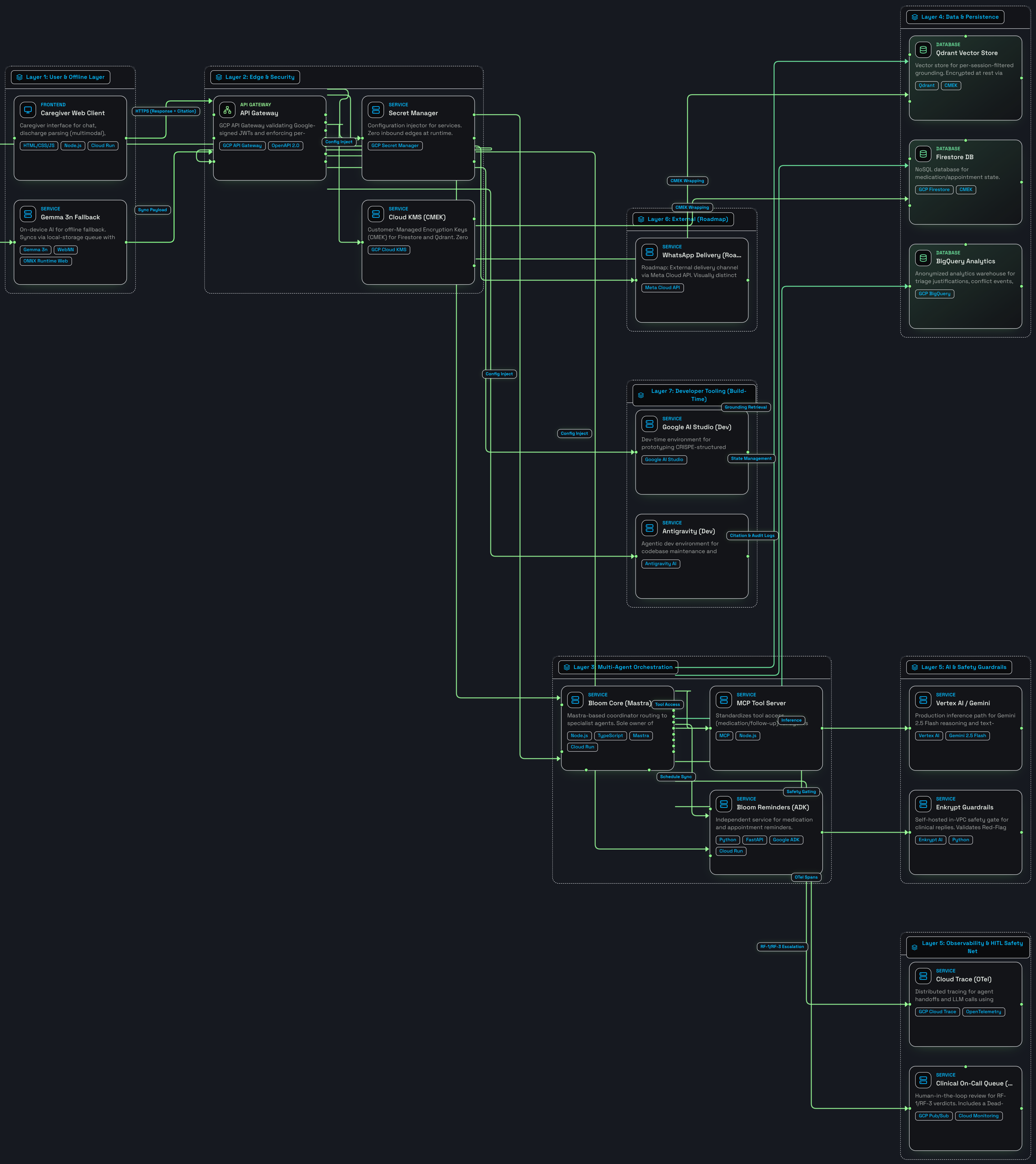
Layer 1: User & Offline Layer
FRONTEND
Caregiver Web Client
Caregiver interface for chat, discharge parsing (multimodal),
HTML/CSS/JS
Node.js
Cloud Run
SERVICE
Gemma 3n Fallback
On-device AI for offline fallback. Syncs via local-storage queue with
Gemma 3n
WebNN
ONNX Runtime Web
HTTPS (Response + Citation)
Sync Payload
Layer 2: Edge & Security
API GATEWAY
API Gateway
GCP API Gateway validating Google-signed JWTs and enforcing per-
GCP API Gateway
OpenAPI 2.0
Config Inject
SERVICE
Secret Manager
Configuration injector for services. Zero inbound edges at runtime.
GCP Secret Manager
SERVICE
Cloud KMS (CMEK)
Customer-Managed Encryption Keys (CMEK) for Firestore and Qdrant. Zero
GCP Cloud KMS
CMEK Wrapping
CMEK Wrapping
Layer 6: External (Roadmap)
SERVICE
WhatsApp Delivery (Roa...
Roadmap: External delivery channel via Meta Cloud API. Visually distinct
Meta Cloud API
Layer 4: Data & Persistence
DATABASE
Qdrant Vector Store
Vector store for per-session-filtered grounding. Encrypted at rest via
Qdrant
CMEK
DATABASE
Firestore DB
NoSQL database for medication/appointment state.
GCP Firestore
CMEK
DATABASE
BigQuery Analytics
Anonymized analytics warehouse for triage justifications, conflict events,
GCP BigQuery
Config Inject
Config Inject
Layer 7: Developer Tooling (Build-Time)
Grounding Retrieval
SERVICE
Google AI Studio (Dev)
Dev-time environment for prototyping CRISPE-structured
Google AI Studio
State Management
SERVICE
Antigravity (Dev)
Agentic dev environment for codebase maintenance and
Antigravity AI
Citation & Audit Logs
Layer 3: Multi-Agent Orchestration
SERVICE
Bloom Core (Mastra)
Mastra-based coordinator routing to specialist agents. Sole owner of
Node.js
TypeScript
Mastra
Cloud Run
Tool Access
SERVICE
MCP Tool Server
Standardizes tool access (medication/follow-up)
MCP
Node.js
Inference
Schedule Sync
Safety Gating
SERVICE
Bloom Reminders (ADK)
Independent service for medication and appointment reminders.
Python
FastAPI
Google ADK
Cloud Run
OTel Spans
Layer 5: AI & Safety Guardrails
SERVICE
Vertex AI / Gemini
Production inference path for Gemini 2.5 Flash reasoning and text-
Vertex AI
Gemini 2.5 Flash
SERVICE
Enkrypt Guardrails
Self-hosted in-VPC safety gate for clinical replies. Validates Red-Flag
Enkrypt AI
Python
RF-1/RF-3 Escalation
Layer 5: Observability & HITL Safety Net
SERVICE
Cloud Trace (OTel)
Distributed tracing for agent handoffs and LLM calls using
GCP Cloud Trace
OpenTelemetry
SERVICE
Clinical On-Call Queue (...
Human-in-the-loop review for RF-1/RF-3 verdicts. Includes a Dead-
GCP Pub/Sub
Cloud Monitoring